

Post-LASIK Tear Dysfunction and Dysesthesia

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Symptoms of tear dysfunction after laser in situ keratomileusis (LASIK) occur in nearly all patients and resolve in the vast majority. Although dry eye complaints are a leading cause of patient discomfort and dissatisfaction after LASIK, the symptoms are not uniform, and the disease is not a single entity. Post-LASIK tear dysfunction syndrome or dry eye is a term used to describe a spectrum of disease encompassing transient or persistent post-operative neurotrophic disease, tear instability, true aqueous tear deficiency, and neuropathic pain states. Neural changes in the cornea and neuropathic causes of ocular surface discomfort may play a separate or synergistic role in the development of symptoms in some patients. Most cases of early post-operative dry eye symptoms resolve with appropriate management, which includes optimizing ocular surface health before and after surgery. Severe symptoms or symptoms persisting after 9 months rarely respond satisfactorily to traditional treatment modalities and require aggressive management. This review covers current theories of post-LASIK dry eye disease, pathophysiology, risk factors, and management options for this disease spectrum of post-LASIK tear dysfunction and neuropathic pain.